Secure Microfinance and Welfare Management Platform Using Blockchain and AI

R26-IT-017

Project Proposal Report

Tharushika A J M H – IT22194626

B.Sc. (Hons) Degree in Information Technology Specialized in Information Technology

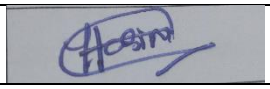
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Sri Lanka Institute of Information Technology Sri Lanka

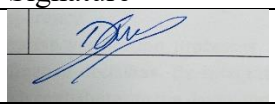
May 2026

DECLARATION

I declare that this is my own work, and this proposal does not knowingly incorporate any material previously submitted for a degree or diploma at any other university or higher education institution, nor does it contain any material that has been previously published or written by another person apart from where proper acknowledgment is given in the text.

Name	Student ID	Signature
Tharushika A J M H	IT22194626	

The above candidate is conducting research for their undergraduate Dissertation under my supervision.

Name	Date	Signature
Dr.Dharshana Kasthurirathna	13th March 2026	

ABSTRACT

Microfinance and welfare programs such as Samurdhi and Aswesuma play a significant role in poverty alleviation and rural development in Sri Lanka. These programs support low-income communities by providing financial assistance, loans, and social welfare benefits. However, the current management of microfinance and welfare services at the Grama Niladhari (GN) division level is largely dependent on manual and fragmented administrative processes. Beneficiary registration, eligibility verification, loan approval, and monitoring activities are often handled through paper-based systems, which result in data duplication, delayed service delivery, biased decision-making, and increased opportunities for fraud and corruption. Although several digital systems and blockchain-based microfinance platforms have been proposed in recent research, most of these solutions primarily focus on recording financial transactions or improving transparency in loan disbursement. Limited attention has been given to automating the eligibility verification process for welfare and microfinance programs using secure decentralized mechanisms. The lack of integrated systems that combine automated rule-based decision making with transparent data governance remains a key research gap. This research proposes a Smart Contract and dApp-based Eligibility Automation System as part of a blockchain-integrated microfinance and welfare management platform. The proposed component utilizes blockchain technology and smart contracts to automatically evaluate eligibility criteria such as income level, welfare history, repayment behavior, and duplicate claims. A decentralized application (dApp) will provide an interface for residents, GN officers, Samurdhi officers, and microfinance institutions to submit applications and review eligibility decisions. All evaluation results and verification actions will be recorded on a blockchain ledger to ensure transparency, accountability, and tamper-resistant records. The expected outcome of this research is a prototype system capable of automating eligibility decisions, reducing administrative workload, preventing fraudulent claims, and improving the efficiency of welfare and microfinance service delivery. The system also demonstrates the potential of blockchain-based governance platforms to enhance transparency, trust, and inclusive digital public services in Sri Lanka.

ACKNOWLEDGEMENT

I would like to express my sincere gratitude to my research supervisor Dr. Dharshana Kasthurirathna for his valuable guidance, continuous support, and constructive feedback throughout the development of this research proposal. His expert knowledge, encouragement, and insightful suggestions greatly helped in shaping the direction and quality of this study.

I would also like to extend my sincere appreciation to my co-supervisor Ms. Chathurya Kumarapperuma for her guidance, support, and helpful advice during the preparation of this research work. Her valuable comments and encouragement significantly contributed to improving the clarity and effectiveness of this proposal.

I would like to thank the academic staff of the Sri Lanka Institute of Information Technology (SLIIT) for providing the necessary knowledge, academic resources, and learning environment that supported the successful completion of this proposal.

My sincere thanks also go to my group members and colleagues for their encouragement, cooperation, and valuable discussions throughout the research process.

Finally, I would like to express my deepest gratitude to my family for their continuous support, patience, and motivation throughout my academic journey. Their encouragement has been a constant source of inspiration in completing this research proposal.

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LIST OF ABBREVIATIONS

Abbreviation	**Full Form**
AI	Artificial Intelligence
API	Application Programming Interface
DSR	Design Science Research
dApp	Decentralized Application
DeFi	Decentralized Finance
GN	Grama Niladhari
IEEE	Institute of Electrical and Electronics Engineers
IT	Information Technology
NLP	Natural Language Processing
SDG	Sustainable Development Goal
UNDP	United Nations Development Programme
WBS	Work Breakdown Structure

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INTRODUCTION

Background and Context

Microfinance and welfare programs play a crucial role in poverty alleviation and socio-economic development in many developing countries. In Sri Lanka, programs such as Samurdhi, Aswesuma, and community-based microfinance initiatives provide financial assistance and small-scale loans to low-income households, particularly in rural communities. These programs are administered primarily at the Grama Niladhari (GN) division level, where government officers manage beneficiary registration, eligibility verification, and service delivery. Research has shown that microfinance initiatives significantly contribute to household income improvement and women's economic empowerment in rural areas of Sri Lanka. However, the effective management of these programs remains challenging due to limitations in existing administrative systems.

Currently, the management of microfinance and welfare services relies heavily on manual and paper-based processes. Beneficiary applications, household data verification, and eligibility decisions are often recorded through fragmented documentation systems maintained separately by GN offices, Samurdhi offices, and microfinance institutions. These disconnected processes lead to delays in decision-making, duplication of records, and difficulties in verifying applicant eligibility. Government officers are required to manually evaluate multiple socio-economic factors such as household income, welfare history, and repayment records, which significantly increases administrative workload and reduces operational efficiency.

The lack of digital integration also creates governance challenges. Manual systems make it difficult to detect duplicate welfare claims, data manipulation, and fraudulent activities, which can result in the misallocation of financial assistance. In addition, limited transparency in eligibility verification processes can reduce public trust in welfare distribution mechanisms. Residents in rural areas also face accessibility challenges due to language barriers, low digital literacy, and the need for repeated visits to administrative offices to complete application procedures.

Although several digital welfare management systems and blockchain-based financial platforms have been proposed in recent research, most existing solutions primarily focus on record management or financial transaction tracking. They rarely provide automated

mechanisms for evaluating eligibility criteria or integrating decision-making processes across multiple stakeholders. In particular, many systems lack rule-based automation capabilities that can consistently evaluate eligibility conditions such as income level, welfare dependency, repayment history, and duplicate claims.

To address these limitations, this research proposes a Smart Contract and dApp-based Eligibility Automation System as a component of a blockchain-integrated microfinance and welfare management platform. The proposed system utilizes smart contracts to automatically evaluate predefined eligibility criteria, enabling transparent and tamper-resistant decision-making. By recording eligibility verification results on a blockchain ledger and providing a decentralized application (dApp) interface for residents and officers, the system aims to reduce administrative workload, improve transparency, and prevent fraudulent claims.

This research aligns with several United Nations Sustainable Development Goals (SDGs), particularly SDG 1: No Poverty, SDG 8: Decent Work and Economic Growth, and SDG 9: Industry, Innovation, and Infrastructure, by supporting inclusive financial services and promoting digital innovation in rural governance systems. Furthermore, the study contributes to the Artificial Intelligence and Blockchain research cluster, focusing on the application of decentralized technologies and intelligent systems to improve transparency, accountability, and efficiency in public-sector financial management.

Literature Review

Recent advancements in blockchain and digital governance technologies have introduced new opportunities to improve transparency, automation, and accountability in financial and welfare management systems. This section reviews existing research related to smart contracts for financial automation, digital welfare management systems, and digital governance platforms, highlighting their contributions and limitations in relation to automated eligibility verification.

Smart contracts are self-executing programs deployed on blockchain networks that automatically enforce predefined rules when specific conditions are met. These contracts

eliminate the need for intermediaries and improve transparency in financial transactions. Swan (2015) explains [1] that blockchain-based smart contracts enable automated execution of financial agreements without relying on centralized authorities, which can significantly reduce operational costs and improve trust among stakeholders. Similarly, Christidis and Devetsikiotis (2016) [2] discuss how blockchain and smart contracts can transform financial services by enabling decentralized automation of transactions and verification processes. Their work highlights the potential of smart contracts in financial applications such as loan management and digital identity systems. Szabo (1997), [3] who originally introduced the concept of smart contracts, emphasized their ability to automate contractual processes through digital protocols, reducing the need for manual enforcement mechanisms. In the context of decentralized finance (DeFi), Buterin (2014) [4] introduced Ethereum, a blockchain platform designed to support programmable smart contracts and decentralized applications (dApps). Ethereum enables developers to build automated financial systems capable of executing complex logic without centralized control. Despite these advancements, many existing blockchain-based financial systems primarily focus on automating financial transactions and loan repayments rather than automating eligibility verification processes. This indicates a gap in the application of smart contracts for rule-based decision-making in welfare and microfinance systems.

Governments around the world have adopted digital systems to manage welfare distribution and beneficiary records. These systems aim to improve efficiency, reduce administrative workload, and increase transparency in public service delivery. The United Nations Development Program (UNDP, 2020) [5] highlights that digital welfare management platforms can significantly improve service delivery by enabling centralized databases for beneficiary information and welfare disbursement tracking. Similarly, Janssen and Estevez (2013) [6] discuss the role of digital government platforms in improving public administration and citizen services through the integration of digital information systems. In many countries, welfare management platforms such as India's Aadhaar-enabled welfare system and digital social protection systems in Europe have improved the accuracy of beneficiary identification and payment distribution. However, most digital welfare systems rely on centralized databases, which introduces several limitations. Centralized systems remain vulnerable to data manipulation, lack transparency in decision-making processes, and often require manual verification of eligibility criteria by administrative officers. Additionally, interoperability between multiple government institutions remains a challenge, particularly in rural administrative environments. These limitations indicate the need for more

secure and transparent digital platforms capable of automating eligibility verification processes.

Digital governance platforms aim to improve transparency, accountability, and efficiency in public administration using digital technologies. Tapscott and Tapscott (2016) [7] argue that blockchain technology has the potential to transform government services by creating transparent and tamper-resistant public records. Blockchain-based governance platforms can enable secure sharing of data across institutions while maintaining data integrity. Atzori (2017) [8] further discusses the concept of blockchain-based governance, explaining how decentralized systems can improve public trust by enabling transparent and verifiable decision-making processes. Although these digital governance platforms demonstrate strong potential for improving transparency and accountability, many existing implementations focus primarily on secure record management and data sharing rather than automated policy enforcement. In particular, few systems integrate smart contracts to automatically evaluate welfare or financial eligibility criteria within decentralized governance platforms.

Based on the reviewed literature, several limitations can be identified in current research and existing systems: Most blockchain-based financial systems focus on transaction transparency and repayment management rather than automating eligibility decision-making. Digital welfare management platforms primarily rely on centralized databases, which remain vulnerable to data manipulation and lack transparent audit mechanisms. Existing governance systems often require manual verification of eligibility criteria, increasing administrative workload and delaying decision-making processes. There is limited research on integrating smart contract–based eligibility automation with welfare and microfinance management at the local administrative level. These limitations highlight the need for a secure and automated eligibility verification system capable of evaluating welfare and microfinance applications using predefined rules while ensuring transparency and accountability.

Research Gap

Although several studies have explored blockchain technologies, digital welfare platforms, and decentralized applications in financial systems, significant limitations remain in the application of these technologies for automated eligibility verification in microfinance and welfare governance systems.

• Most blockchain-based financial systems focus mainly on recording financial transactions such as loan disbursement and repayment, rather than automating eligibility evaluation for welfare and microfinance programs.

• Many digital welfare management platforms rely on centralized databases, which lack transparency and remain vulnerable to data manipulation and fragmented data management across institutions.

• Existing decentralized applications (dApps) in blockchain ecosystems are primarily designed for cryptocurrency-based financial services such as decentralized lending, and they do not support policy-based eligibility verification using socio-economic criteria.

• Current systems rarely provide multi-stakeholder governance mechanisms that allow government officers, welfare authorities, and microfinance institutions to collaboratively verify eligibility decisions.

• There is a lack of integrated platforms that combine smart contract–based rule enforcement with a decentralized application (dApp) interface to automate eligibility evaluation while ensuring transparent and tamper-resistant decision records.

Therefore, there is a clear need for a Smart Contract + dApp Eligibility Automation System that can automatically evaluate welfare and microfinance applications using predefined rule-based criteria while providing a decentralized interface for stakeholders to interact with the system.

Technologies and Methods	Research Paper [1] Blockchain Microfinance	Research Paper [2] Smart Contract Finance	Research Paper [3] Digital Welfare Systems	Research Paper [4] Blockchain Governance	Research Paper [5] DeFi dApps	Proposed System
Blockchain-based Transaction Recording	✅	✅	❌	✅	✅	✅

Smart Contract Automation	❌	✅	❌	✅	✅	✅
Eligibility Rule Evaluation	❌	❌	✅ (manual)	❌	❌	✅
Multi-stakeholder Verification	❌	❌	✅	❌	❌	✅
Automated Eligibility Decision	❌	❌	❌	❌	❌	✅
Decentralized Application (dApp) Interface	❌	✅	❌	❌	✅	✅

Table 1 Comparison of the existing research methods and proposed method

From the comparison above, it is evident that existing systems address only individual aspects of microfinance and welfare management, such as transaction transparency, financial automation, or digital record management. However, none of the existing approaches provide a comprehensive solution that integrates smart contract–based eligibility automation with a decentralized application interface and multi-stakeholder governance mechanisms.

The proposed Smart Contract + dApp Eligibility Automation System addresses this gap by enabling automated eligibility verification using rule-based smart contracts, while allowing residents, government officers, and financial institutions to interact with the system through a decentralized application interface. This approach ensures transparent, tamper-resistant, and efficient decision-making for microfinance and welfare management.

OBJECTIVE

Main Objective

The overarching objective of this research is to design, develop, and evaluate a blockchain-based Smart Contract and decentralized application (dApp) eligibility automation system that enables transparent, secure, and rule-based verification of microfinance and welfare applications by automatically enforcing predefined eligibility criteria and recording decisions on a tamper-resistant distributed ledger.

This objective aims to address the limitations of manual eligibility verification processes in rural governance systems by introducing an automated, transparent, and decentralized mechanism for evaluating welfare and microfinance applications.

Specific Objectives

1. Investigate existing microfinance and welfare eligibility verification processes

    This objective examines current administrative procedures used in microfinance and welfare management at the GN division level. The analysis focuses on identifying inefficiencies in manual eligibility verification, risks of fraud and duplicate claims,

fragmented data management across institutions, and the overall limitations of existing paper-based processes.

2. Model a rule-based eligibility evaluation framework

   This objective defines and structures a set of eligibility criteria for welfare and microfinance programs using socio-economic indicators such as household income level, welfare dependency history, repayment behavior, and duplicate claim detection. The framework will provide the decision logic required for automated eligibility evaluation.

3. Design and implement smart contracts for automated eligibility verification

   This objective focuses on developing blockchain-based smart contracts that enforce predefined eligibility rules. The smart contracts will process applicant data and generate transparent approval or rejection decisions while eliminating reliance on manual decision-making processes.

4. Develop a decentralized application (dApp) for multi-stakeholder interaction

   This objective involves designing and implementing a decentralized application that enables residents, Grama Niladhari officers, Samurdhi officers, and microfinance institutions to submit applications, review applicant data, and interact with the eligibility verification system through a secure and user-friendly interface.

5. Integrate the smart contract system with blockchain infrastructure and the dApp interface

   This objective establishes seamless communication between the decentralized application and the smart contract layer. The integration enables real-time eligibility evaluation and ensures that verification decisions are securely recorded on the blockchain ledger.

6. Evaluate the performance and effectiveness of the eligibility automation system

   This objective evaluates the proposed system using experimental testing with sample datasets. The evaluation focuses on measuring improvements in transparency, efficiency of eligibility verification, fraud prevention capability, and reduction of administrative workload compared with traditional manual processes.

METHODOLOGY

Research Approach

This study adopts a Design Science Research (DSR) methodology, which is widely used in information systems research to design and evaluate technological solutions that address real-world problems. The purpose of this research is to develop and evaluate a Smart Contract and decentralized application (dApp)–based eligibility automation system that improves transparency and efficiency in microfinance and welfare management.

The Design Science approach is appropriate because the research focuses on the creation of an innovative artifact, namely the eligibility automation platform that integrates blockchain smart contracts and a decentralized application interface. The research process consists of several stages: problem identification, system design, artifact development, implementation, and evaluation. Initially, existing microfinance and welfare verification processes will be analyzed to identify limitations in manual decision-making and fragmented data management. Based on this analysis, a rule-based eligibility evaluation framework will be designed and implemented through blockchain smart contracts. A decentralized application (dApp) will then be developed to enable stakeholders to interact with the system. Finally, the effectiveness of the system will be evaluated using experimental testing with sample datasets.

Technical Methods and Algorithms

The proposed system employs a rule-based decision automation mechanism implemented through blockchain smart contracts. The eligibility verification logic is defined as a set of deterministic rules derived from socio-economic eligibility criteria used in welfare and microfinance programs.

The system processes applications through the following workflow:

- Application Submission

  Residents submit welfare or microfinance applications through the decentralized application (dApp). The application includes socio-economic data such as income level, welfare history, household size, and repayment records.

- Data Verification

  Authorized stakeholders such as Grama Niladhari officers or Samurdhi officers review and verify the submitted information before the eligibility evaluation process begins.

- Rule-Based Eligibility Evaluation

  Smart contracts automatically evaluate applicant data by applying predefined eligibility rules. The rule-based algorithm compares the submitted data with policy-defined thresholds and conditions to determine eligibility status.

- Blockchain Recording

  Once the smart contract executes the eligibility evaluation, the decision and verification details are recorded on the blockchain ledger to ensure transparency and immutability.

- Decision Notification

  The dApp interface displays the eligibility decision to authorized users and provides an auditable record of the evaluation process.

This rule-based smart contract mechanism ensures consistent, transparent, and tamper-resistant eligibility decisions.

Tools and Platforms

The proposed system will be developed using blockchain technologies and modern web development frameworks suitable for decentralized applications.

- Blockchain Platform –Hyperledger Fabric

  A blockchain platform will be used to deploy smart contracts and store eligibility verification results on an immutable ledger. Ethereum supports decentralized applications and smart contracts, while Hyperledger Fabric provides a permissioned blockchain suitable for government or institutional systems.

- Smart Contract Development – Solidity

Smart contracts will be developed using Solidity, a programming language designed for creating and executing smart contracts on Ethereum-based blockchain networks.

- Decentralized Application Development – React.js and Web3.js

  The decentralized application (dApp) will be developed using React.js for the user interface and Web3.js to enable interaction between the application and blockchain smart contracts.

- Smart Contract Development Framework – Hardhat / Truffle

  Hardhat or Truffle will be used to compile, test, and deploy smart contracts during development.

HIGH-LEVEL SYSTEM ARCHITECTURE

Overview of the Proposed System

The proposed system is a Smart Contract and decentralized application (dApp)–based eligibility automation platform designed to support transparent and efficient microfinance and welfare management.

The system automates eligibility verification by applying predefined rule-based criteria using blockchain smart contracts. A decentralized application (dApp) provides a user interface for residents, government officers, and microfinance institutions to submit applications and review eligibility decisions.

The architecture consists of three main layers:

• User Interface Layer (dApp) – enables users to submit and review applications.
• Smart Contract Layer – enforces eligibility rules and processes application data.
• Blockchain Data Layer – stores verification records and eligibility decisions securely.

High-Level System Components

The proposed system consists of the following key components:

1. Decentralized Application (dApp)

Provides the interface for residents and government officers to interact with the system, submit applications, and view eligibility results.

2. Smart Contract Eligibility Engine

Implements rule-based logic that automatically evaluates applicant data and determines eligibility.

3. Blockchain Network

Stores verification records and eligibility decisions on an immutable distributed ledger to ensure transparency and data integrity.

System Architecture Diagram

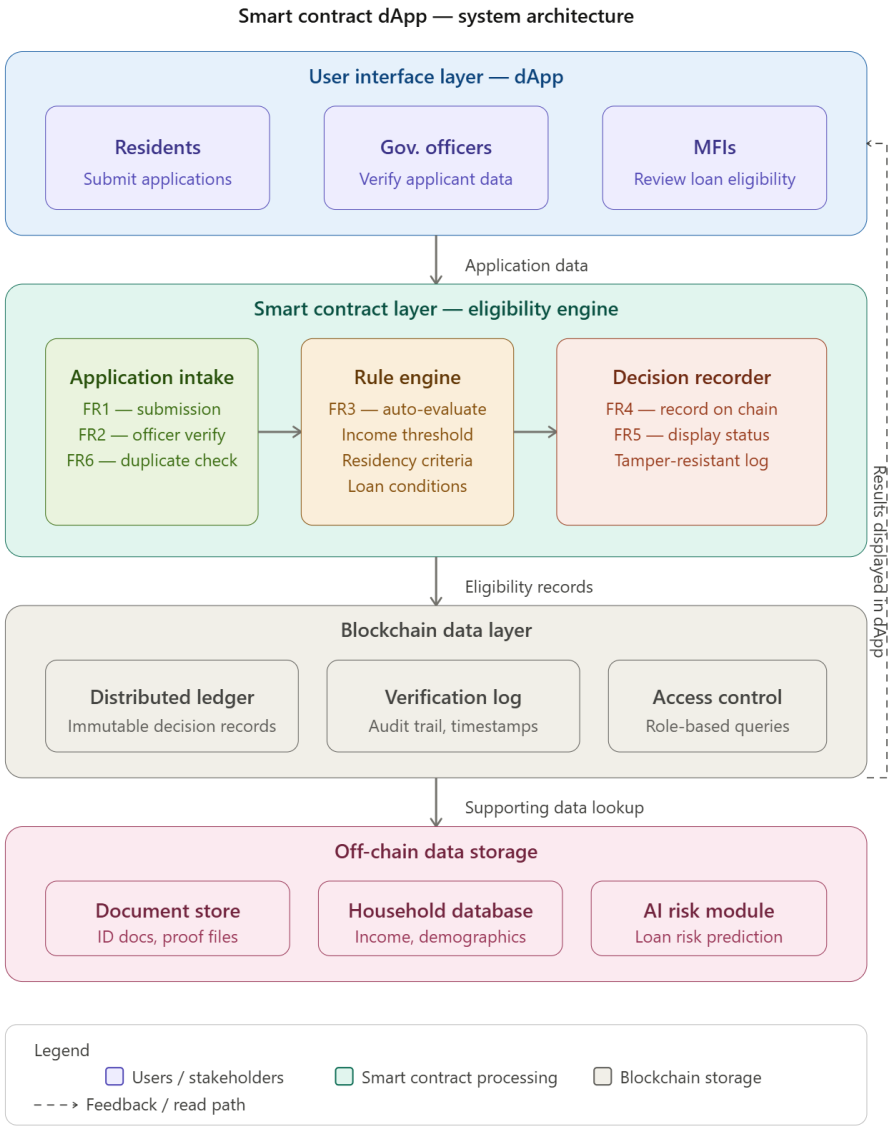


Figure 1 - High-level System Architecture

Component Overview

The Smart Contract + dApp Eligibility Automation Component is responsible for automating the verification of welfare and microfinance applications using blockchain-based smart contracts. The component integrates a decentralized application (dApp), a smart contract eligibility engine, and a blockchain network to ensure transparent and tamper-resistant decision-making.

Component Interaction and Data Flow

- The component operates through coordinated interactions between users, the decentralized application, and blockchain smart contracts.
- Residents submit welfare or microfinance applications through the dApp interface, providing required socio-economic information such as income level and welfare history.
- Authorized officers (e.g., Grama Niladhari or Samurdhi officers) review and verify the submitted information through the system.
- Once verified, the application data is transmitted to the smart contract eligibility engine, where predefined eligibility rules are executed automatically.
- The smart contract evaluates the applicant data using rule-based logic and determines the eligibility status.
- The eligibility decision is recorded on the blockchain ledger, ensuring transparency, immutability, and auditability of the decision-making process.
- The eligibility results are then displayed to authorized stakeholders through the dApp interface, allowing them to review the outcome and application status.
- This automated workflow ensures that eligibility verification is conducted consistently, efficiently, and without manual bias.

Scalability and Security Considerations

- The proposed component is designed with scalability and security in mind to support reliable system operation.

Scalability

- Large or sensitive data that does not require blockchain immutability is stored in an off-chain database to reduce blockchain storage overhead.

- The modular system architecture allows integration with other components of the overall platform, such as AI-based loan risk prediction and multilingual NLP services.

Security

- Blockchain technology ensures immutability and transparency of eligibility verification records.
- Smart contracts enforce rule-based decision automation, reducing human errors and opportunities for manipulation.
- Sensitive user information is protected using data anonymization and secure access control mechanisms to preserve privacy.
- Through this design, the component provides a scalable, secure, and transparent solution for automating eligibility verification in microfinance and welfare management systems.

USER REQUIREMENTS

The system involves several key stakeholders:

- Residents – individuals applying for welfare assistance or microfinance loans.
- Grama Niladhari Officers – responsible for verifying household and applicant information.
- Samurdhi Officers – responsible for managing welfare program eligibility.
- Microfinance Institutions – evaluate loan applications and repayment conditions.
- System Administrators – maintain system functionality and manage access permissions.

Functional Requirements

- FR1 – The system shall allow residents to submit welfare or microfinance applications through the dApp.
- FR2 – The system shall allow authorized officers to verify applicant information.
- FR3 – The system shall automatically evaluate eligibility criteria using smart contracts.
- FR4 – The system shall record eligibility decisions on the blockchain ledger.

- FR5 – The system shall allow stakeholders to view application status and eligibility results.
- FR6 – The system shall detect duplicate applications or conflicting records.

Non-Functional Requirements

Security

- The system shall ensure data integrity using blockchain technology.
- Access to sensitive information shall be restricted to authorized users.

Performance

- Eligibility verification should be processed within a reasonable response time.

Usability

- The dApp interface should be simple and user-friendly for both residents and officers.

Reliability

- The system should maintain accurate and tamper-resistant records of eligibility decisions.

COMMERCIALIZATION PLAN

The proposed Smart Contract and decentralized application (dApp)–based eligibility automation system is mainly intended for organizations involved in welfare and microfinance management, such as government agencies, microfinance institutions, and local administrative offices like Grama Niladhari divisions. The system helps these organizations improve the efficiency and transparency of eligibility verification processes. It provides value by automating eligibility checks using smart contracts, reducing manual work for officers, preventing fraudulent claims, and maintaining transparent and tamper-resistant records through blockchain technology. The system can generate revenue through software licensing for government organizations, subscription-based services for financial institutions, and customization or implementation support. The main costs include system development, blockchain or cloud infrastructure, and deployment or training expenses. Pricing can be based

on licensing agreements, subscription plans, or partnerships with government agencies. The system also offers a competitive advantage compared to existing solutions by combining smart contract automation, blockchain transparency, and a user-friendly decentralized application interface. In addition, the system design and smart contract logic can be protected through software copyright and intellectual property registration.

BUDGET AND JUSTIFICATION

Resources	Cost (LKR)
Internet	4,000
Electricity	5,000
Transport	7,000
Communication	2,000
Purchasing	10,000
Paper Publish Cost	5,000
Total Estimated Budget	33,000

Table 2 - Budget Estimated

Justification for Expenses

Internet (LKR 4,000)

Internet access is essential for conducting literature reviews, accessing online research databases, downloading blockchain development tools, and maintaining communication with team members and supervisors.

Electricity (LKR 5,000)

Electricity is required for operating computers, running blockchain development environments, testing system components, and maintaining continuous development work.

Transport (LKR 7,000)

Transportation expenses are associated with traveling for project meetings, research discussions, and consultations with supervisors or team members.

Communication (LKR 2,000)

Communication costs include phone calls, online meetings, and collaboration with team members during the research and development process.

Purchasing (LKR 10,000)

This budget covers purchasing essential development resources such as hardware accessories, development tools, or software resources required during the project.

Paper Publish Cost (LKR 5,000)

This cost is allocated for publishing the research paper in academic conferences or journals to present the research findings and contribute to academic knowledge.

WORK BREAKDOWN STRUCTURE (WBS)

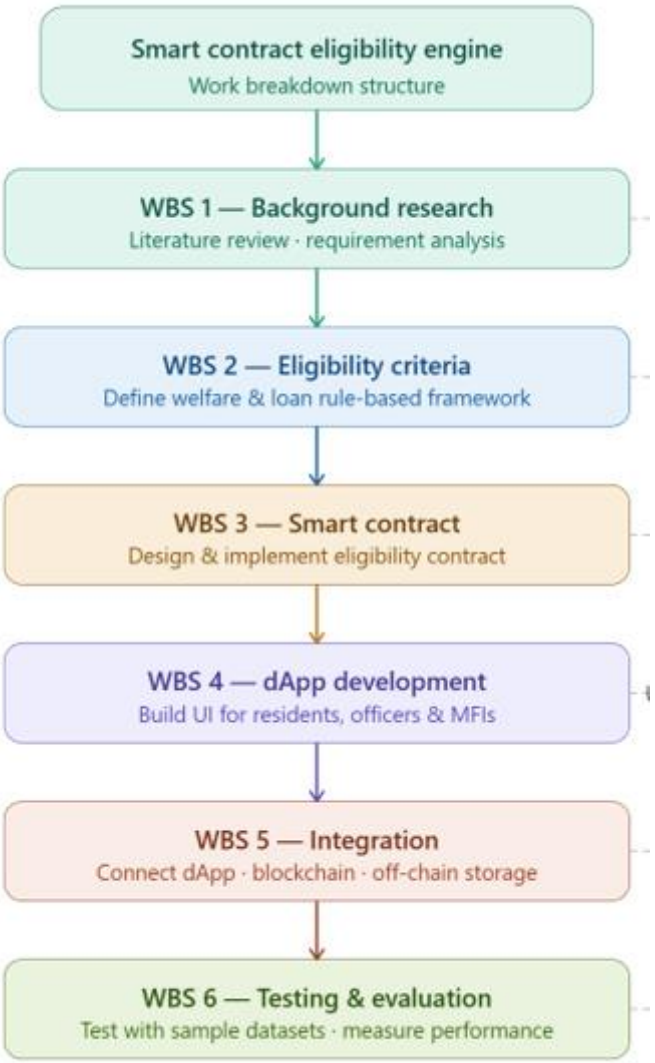


Figure 2-Work Breakdown Structure

GANTT CHART

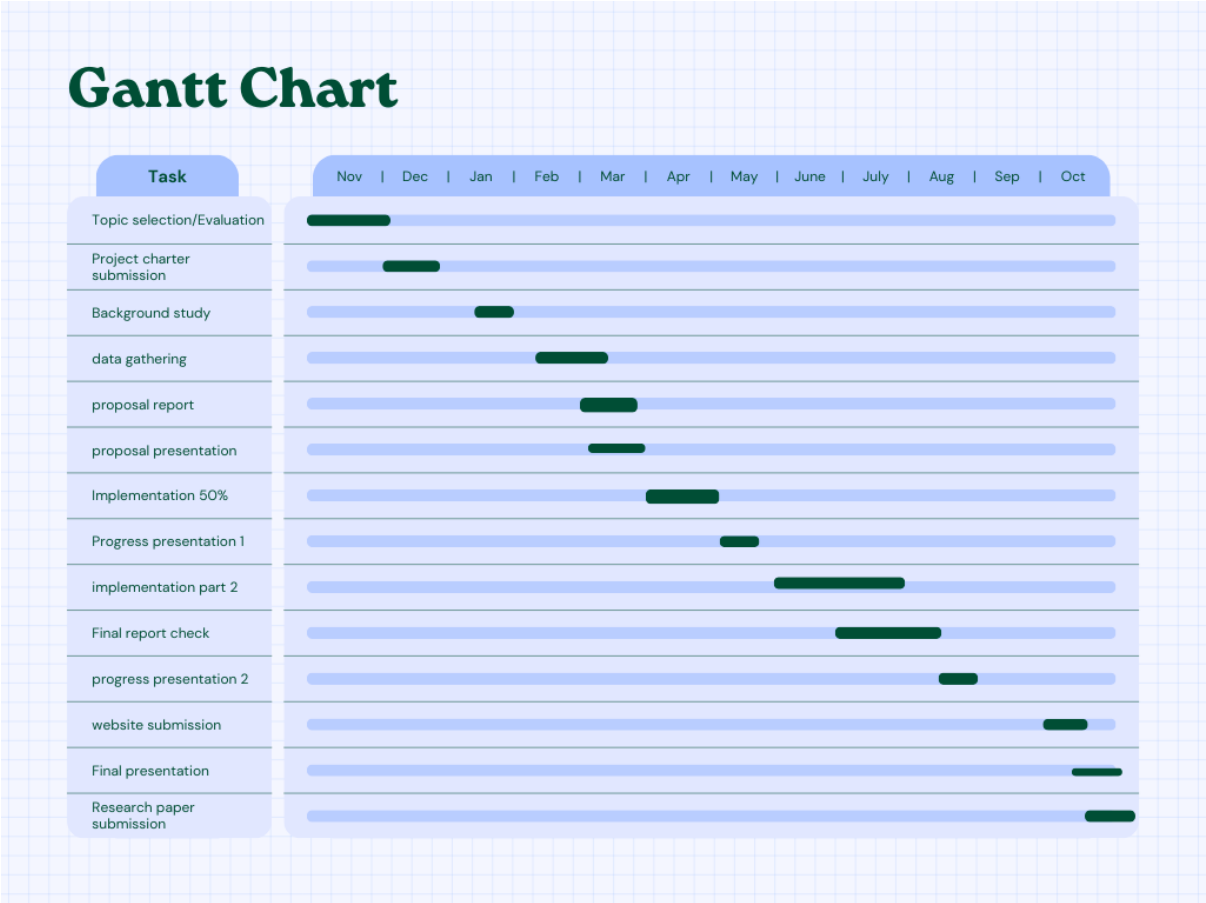


Figure 3-Gantt Chart

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APPENDICES

Appendix A

Loan Application Data Attributes (Provided by Samupakara Gramiya Sanwardhana Bank)

Sample Ledger Portfolio (Microfinance Credit Scoring)

Field	Details / Example	Purpose
Customer ID / Account No.	2025-MF-001	Unique identifier
Customer Name	John Silva	Identification
National ID / Passport No.	123456789V	Verification
Age / Gender	35 / Male	Demographics
Address / Location	Colombo, Sri Lanka	Regional mapping
Household Size	4	Dependency ratio
Loan Type	Gold-backed / Micro loan	Product category
Loan Amount	LKR 50,000	Exposure
Collateral Details	22K Gold chain, 25g	Security
Monthly Income	LKR 35,000	Repayment capacity
Monthly Expenses	LKR 20,000	Debt-to-income ratio
Savings / Deposits	LKR 5,000	Financial discipline
Existing Loans	LKR 15,000 (other MFI)	Over-indebtedness check
Repayment History	95% on-time	Reliability
Attendance (Group Meetings)	90%	Commitment
Mobile Money Usage	Active (weekly transactions)	Alternative data
Utility Bill Payments	On-time	Responsibility
Risk Flags	None	Fraud detection
Credit Score	78/100 (Band B)	Risk rating
Decision	Approved – Conditional	Loan eligibility
Officer Remarks	Customer reliable, collateral verified	Governance
Date of Entry	08 Dec 2025	Audit trail

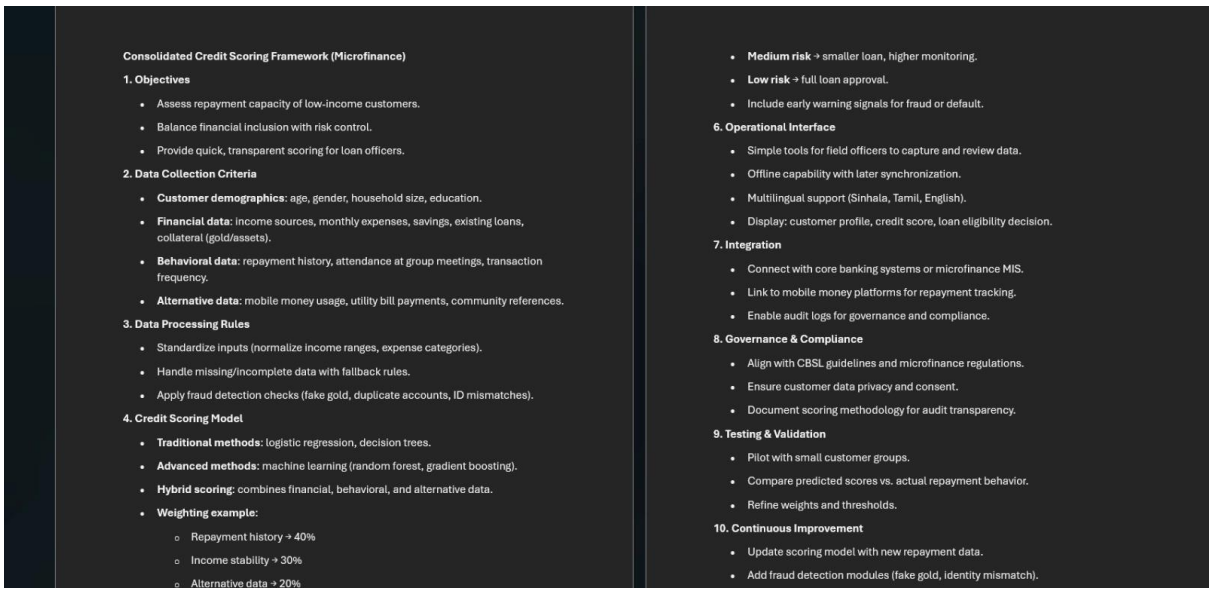
✅ **Key Notes**

- This **ledger portfolio** can be digitized into your app database.
- Each row = one customer record.
- You can add **columns for audit logs** (who entered, who approved).

Appendix 1 - Loan Application Data Attributes (Provided by Samupakara Gramiya Sanwardhana Bank)

Appendix B

Bank Requirement Document



Consolidated Credit Scoring Framework (Microfinance)

1. Objectives

- Assess repayment capacity of low-income customers.
- Balance financial inclusion with risk control.
- Provide quick, transparent scoring for loan officers.

2. Data Collection Criteria

- **Customer demographics**: age, gender, household size, education.
- **Financial data**: income sources, monthly expenses, savings, existing loans, collateral (gold/assets).
- **Behavioral data**: repayment history, attendance at group meetings, transaction frequency.
- **Alternative data**: mobile money usage, utility bill payments, community references.

3. Data Processing Rules

- Standardize inputs (normalize income ranges, expense categories).
- Handle missing/incomplete data with fallback rules.
- Apply fraud detection checks (fake gold, duplicate accounts, ID mismatches).

4. Credit Scoring Model

- **Traditional methods**: logistic regression, decision trees.
- **Advanced methods**: machine learning (random forest, gradient boosting).
- **Hybrid scoring**: combines financial, behavioral, and alternative data.
- **Weighting example**:
  - Repayment history → 40%
  - Income stability → 30%
  - Alternative data → 20%
- **Medium risk** → smaller loan, higher monitoring.
- **Low risk** → full loan approval.
- Include early warning signals for fraud or default.

6. Operational Interface

- Simple tools for field officers to capture and review data.
- Offline capability with later synchronization.
- Multilingual support (Sinhala, Tamil, English).
- Display: customer profile, credit score, loan eligibility decision.

7. Integration

- Connect with core banking systems or microfinance MIS.
- Link to mobile money platforms for repayment tracking.
- Enable audit logs for governance and compliance.

8. Governance & Compliance

- Align with CBSL guidelines and microfinance regulations.
- Ensure customer data privacy and consent.
- Document scoring methodology for audit transparency.

9. Testing & Validation

- Pilot with small customer groups.
- Compare predicted scores vs. actual repayment behavior.
- Refine weights and thresholds.

10. Continuous Improvement

- Update scoring model with new repayment data.
- Add fraud detection modules (fake gold, identity mismatch).

Appendix 2 - Bank Requirement Document